

Vector-First Extraction of GC-MS Stick Spectra from PDF Documents

Abstract

Gas chromatography–mass spectrometry (GC-MS) is the most widely relied-upon technique for identifying volatile compounds, and decades of EI-MS measurements have accumulated in the published literature. However, spectra are almost universally shared as figures embedded in PDF documents, making them inaccessible to programmatic reuse; existing digitization tools either require manual intervention or are designed for continuous line plots rather than the discrete bar (stick) spectra that mass spectrometry produces. We present a two-tier pipeline that recovers m/z –intensity peak lists from such figures by reading the PDF drawing (vector) layer directly, preserving the exact bar geometry without rasterization, and falling back to a raster-based image path only when the vector layer is absent. Extracted peaks are refined by a label-anchored calibration step and screened by an automated physics-based validation layer, and a parallel batch driver scales the method to corpus-sized collections. We assess extraction fidelity against a large reference set of NIST library spectra, cross-pipeline consistency on independently produced real-instrument spectra, and parallel performance. Reading the vector layer recovers the published peak patterns with near-lossless fidelity and high throughput, offering a practical route to unlocking the machine-readable spectral data currently trapped in the figures of the literature.

1 Introduction

GC-MS (Gas Chromatography–Mass Spectrometry) is an analytical technique that unites the separation of compounds by gas chromatography with their identification by mass spectrometry. Its applications range from forensic science, drug detection, and environmental analysis to the identification of unknown compounds. Its reliability rests on electron ionization (EI), which breaks each molecule apart in a consistent, instrument-independent way: the resulting pattern of fragment masses behaves like a fingerprint that can be looked up in curated reference libraries. It is this reproducibility that has made GC-MS the benchmark technique for identifying volatile compounds [1].

The volume of chemical data being produced is growing rapidly, yet much of it remains locked in formats that are difficult to process programmatically. PDFs in particular are often described as “dead-end formats” because information

cannot be extracted from them efficiently or accurately [2], and mass spectra are most commonly shared as figures embedded in PDFs. Existing tools such as WebPlotDigitizer are accurate but tedious to use, particularly across large collections of files or on dense spectra [3]. For mass-spectra graphs specifically, an accurate and automatic tool for recovering every data point is still lacking. Moreover, digitizing such figures by rasterization recovers only a subset of the available information [2]; we therefore read the PDF drawing (vector) layer directly rather than its rendered pixels, which preserves the underlying geometry exactly.

With an effective means of extracting this data, machine-learning approaches to the identification and classification of unknown compounds could become substantially more robust, helping to mitigate the incomplete compound coverage of the NIST database [4].

In this paper, we present a pipeline for extracting m/z -versus-intensity data from bar (stick) mass spectra. Our contributions are:

1. A vector-first extraction method that recovers exact m/z -intensity peaks directly from the PDF drawing layer, with no machine learning and no rasterization in the common case.
2. A rasterization-based fallback that handles figures whose vector layer is absent or unusable, so the pipeline degrades gracefully rather than failing.
3. A large-scale round-trip fidelity evaluation comparing $\sim 50,000$ spectra extracted from NIST-rendered PDFs against the same library entries they were rendered from, by cosine similarity; cross-pipeline consistency on independently produced, real-instrument spectra is assessed separately.
4. A parallel batch driver that achieves up to $6.94\times$ speedup and 917 PDFs/s throughput with a process pool (Amdahl parallel fraction 89.7%, theoretical ceiling $9.7\times$).

2 Related Work

The work most relevant to this paper falls into three areas: general-purpose figure digitization tools, mass-spectral library construction and machine learning, and vector-layer extraction from PDF documents.

2.1 Plot and figure digitization

Early approaches to recovering data from scientific figures focused on semi-automatic tools that require substantial manual setup. WebPlotDigitizer [3], the most widely used tool, requires the user to manually align axes, enter tick values, select the colour of the target plot, and draw the extraction region for each file. This per-file overhead is impractical at corpus scale and fails on densely packed or overlapping curves.

The closest fully automatic prior work is Plot2Spectra [5], which digitizes XANES and Raman spectroscopy images using an anchor-free object detector for axis alignment and a semantic segmentation network with optical-flow constraints for line separation. Plot2Spectra was designed for continuous line plots, where data is encoded as an unbroken curve. Bar mass spectra are structurally different: each data point is a discrete vertical stroke at an integer mass, and the per-pixel instance-embedding approach Plot2Spectra uses for multi-line separation is inapplicable to isolated bars. Critically, both Plot2Spectra and Web-PlotDigitizer operate on rasterized images, discarding the exact vector geometry that most PDF-based spectra encode. We target discrete stick spectra and read the PDF vector layer directly, avoiding the quantization loss that rasterization introduces.

2.2 Mass-spectral libraries and machine learning

The NIST/EPA/NIH Mass Spectral Library is the primary reference for EI-MS compound identification [6]. Library search uses the cosine similarity between an experimental spectrum and each reference entry as the match score, a metric we adopt for our correctness evaluation [7]. Despite containing over 350,000 compounds, coverage is incomplete and grows slowly: each update cycle adds approximately 40,000 new spectra [4]. This gap motivates machine-learning approaches that predict mass spectra from molecular structure or use library spectra to train classifiers for compound identification [4]. All such methods require large labelled spectral datasets, which the published literature contains in the form of figures but not in a machine-readable form. Our extractor provides a route to recovering this data at scale, directly addressing the training-data bottleneck.

2.3 Vector-layer processing in PDF documents

The PDF specification defines page content as a stream of exact drawing operators: straight strokes, filled paths, and text positioned in a device-independent coordinate system [8]. Reading these operators directly recovers the renderer’s geometric intent without loss. By contrast, rasterizing a PDF page and re-detecting its geometry with image-processing algorithms introduces pixel quantization, thresholding noise, and DPI-dependent aliasing. The data-liberation literature has noted that recovering numerical values from rasterized figures retrieves only a subset of the encoded information [2]. By targeting the vector layer as the primary extraction path, our pipeline avoids this information loss entirely for the 99.9% of files in our corpus that expose a usable drawing layer, falling back to rasterization only when no vector layer is present.

3 Methods

3.1 Gas Chromatography–Mass Spectrometry

Gas chromatography–mass spectrometry (GC-MS) is a two-stage analytical instrument that separates and identifies the chemical components of a sample. In the first stage, the gas chromatograph vaporises the sample and carries it through a capillary column using an inert carrier gas. Because different compounds travel through the column at different rates—governed by their volatility and affinity for the stationary phase—they arrive at the detector separated in time, one compound eluting per retention-time window. In the second stage, each eluting compound enters the mass spectrometer, where an electron-ionization (EI) source fragments it into charged pieces. A mass analyser (typically a quadrupole) then sorts those fragments by their mass-to-charge ratio (m/z), and the detector records the abundance of each fragment, producing a bar-chart spectrum of m/z versus relative intensity with the tallest peak normalised to 100. Because EI fragmentation is highly reproducible across instruments and laboratories, the resulting spectrum is effectively a chemical fingerprint that can be matched against curated reference libraries for compound identification.

The instrument stores these spectra—one per retention-time peak—as PDF reports. Each report embeds a vector-graphics rendering of the m/z -intensity bar chart, and it is from these embedded charts that this work automatically recovers the numerical peak lists.

3.2 Vector extraction

The following describes how the pipeline reads the PDF drawing layer to recover the exact bar geometry of a mass spectrum, from library selection through axis calibration and peak extraction.

Choice of PDF library. Both tiers are built on PyMuPDF¹ (the `fitz` binding to the MuPDF C library). The two-tier design requires three operations from a single library in a common coordinate system: extraction of the vector drawing operators (`get_drawings`), positioned text extraction for tick-label calibration (`get_text("words")`), and page rendering for the raster fallback (`get_pixmap`). Among widely used Python PDF libraries, only PyMuPDF exposes all three: `pikepdf`, `PyPDF2`, and `pdfcrowd` provide neither vector-graphics extraction nor page rendering, and `pdfplumber`/`pdfminer` provide only limited vector extraction and no rendering [9]. Performance further motivates the choice at corpus scale. On the maintainers’ reference benchmark of 7,031 pages, PyMuPDF extracts text in 8.0s, against 101.6s for `PyPDF2` and 227.3s for `PDFMiner`—a 13–28× advantage—and renders pages in 367s, against 646s for `XPDF` and

¹PyMuPDF and the underlying MuPDF library are dual-licensed under the AGPL and a commercial license [9].



Figure 1: Schematic of a GC-MS instrument. The gas chromatograph (left) separates sample components by retention time; each eluting compound enters the mass spectrometer (right), where EI fragmentation and mass analysis produce the m/z -intensity spectrum stored in the PDF reports processed by this work.

852 s for PDF2JPG [9]. Because per-file parsing lies on the critical path of the parallel driver (§5.5), this margin directly affects achievable throughput.

PDFs store graphical content as a sequence of vector drawing operations. We collect each drawing operation and flatten every straight stroke — both line items and the edges of rectangles — into a set of line segments expressed in the native PDF coordinate system. Because each stroke encodes a shape’s geometry exactly, this recovers the positions of the axes and bars directly, with no rasterization, thresholding, or learned detection, and hence without the quantization error those steps introduce.

Axis and baseline detection. Each spectrum is a 2-D x - y graph, so the first elements to identify are the two axes. The x -axis (the intensity baseline) is taken as the longest near-horizontal segment in the lower half of the page; the y -axis is the leftmost sufficiently long near-vertical segment. Their intersection defines the plot origin. A segment is treated as horizontal when its endpoints differ by less than 0.5 PDF units in y , and as vertical when they differ by less than 0.5 PDF units in x . Writing W and H for the page width and height (with y increasing downward), the x -axis is the longest horizontal segment exceeding $0.3 W$ in length whose y -midpoint lies below the top 30 % of the page; the y -axis is the *leftmost* vertical segment exceeding $0.15 H$ in length whose x -midpoint lies within the left $0.4 W$, with a $0.02 W$ margin excluded at each page edge so that

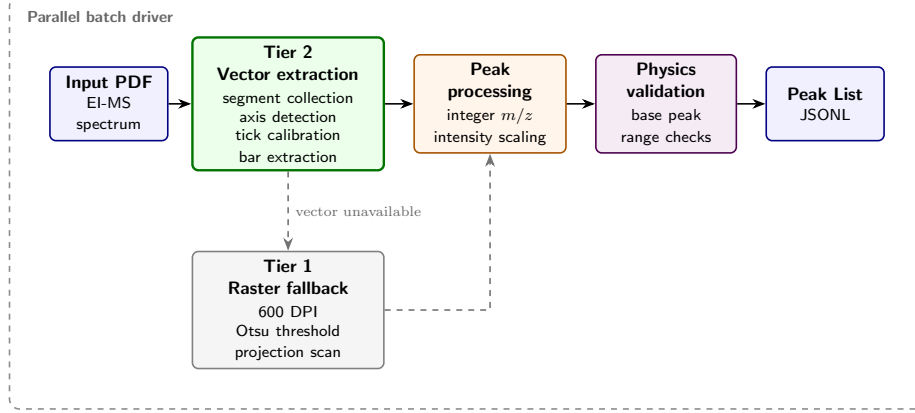


Figure 2: Overview of the extraction pipeline. A PDF is processed by the vector-extraction path; if the vector layer is unavailable, a raster fallback is used. Both paths feed a shared post-processing and validation stage before producing the final m/z -intensity peak list.

a full-page border rectangle is not mistaken for the axis. The leftmost (rather than longest) vertical is taken because a base peak at 100 % can be exactly as tall as the axis. These layout constraints are empirically chosen.

Calibration. Mapping pixel positions to m/z values is performed using the numeric tick labels printed below the x -axis. Each label yields a pair (x_i, v_i) , where x_i is the pixel abscissa of the label’s centre and v_i its parsed numeric value; in the example of Fig. 3 these are the ticks at 20, 40, $\dots$, 320. A least-squares fit over all such pairs gives the linear calibration

$$m/z = \frac{x - x_0}{s_x}, \quad (1)$$

where s_x is the number of pixels per mass unit and x_0 is the pixel abscissa corresponding to $m/z = 0$. The intensity axis is calibrated from the “100” label at the top of the y -axis, which defines the 100 % level, together with the baseline at 0 %:

$$I = \min\left(\frac{h}{s_y}, 100\right), \quad (2)$$

where s_y is the fitted number of pixels per intensity percent and h is a bar’s pixel height above the baseline; the clamp at 100 guards against a stroke that slightly overshoots the labelled maximum.

Bar extraction. With axes and calibration established, candidate bars are the vertical segments that touch the x -axis baseline, lie to the right of the y -axis, and fall within the plot bounds. Each segment’s abscissa and height above the baseline give a provisional m/z and an intensity through Eqs. (1) and (2) respectively; the m/z assignment is refined by the next step. Segments whose abscissae coincide to within a tolerance of 0.01 PDF units are merged into a

Algorithm 1 Tier dispatch for mass spectrum extraction

Require: PDF file f , Boolean flag $cross_validate$ **Ensure:** $\{name, peaks, method, warnings\}$

```
1:  $name \leftarrow \text{EXTRACTCOMPOUNDNAME}(f)$ 
2:  $peaks \leftarrow \emptyset$ ;  $warnings \leftarrow []$ ;  $method \leftarrow \text{VECTOR}$ 
3: Try
4:    $peaks \leftarrow \text{VECTORPIPELINE}(f)$ 
5: Catch ExtractionFailure
6:    $method \leftarrow \text{RASTER}$ 
7:    $peaks \leftarrow \text{RASTERPIPELINE}(f)$ 
8: EndTry
9:  $warnings \leftarrow \text{PHYSICSCHECKS}(peaks)$ 
10: if  $cross\_validate$  and  $method = \text{VECTOR}$  then
11:    $peaks_r \leftarrow \text{RASTERPIPELINE}(f)$ 
12:    $\Delta_n \leftarrow \frac{||peaks| - |peaks_r||}{\max(|peaks|, |peaks_r|)}$ 
13:   if  $\Delta_n > 0.20$  then
14:      $warnings.APPEND(\text{count mismatch})$ 
15:   end if
16:    $M_v \leftarrow \{p.mz \mid p \in peaks\}$ ;  $M_r \leftarrow \{p.mz \mid p \in peaks_r\}$ 
17:    $only\_v \leftarrow \{m \in M_v \mid \nexists m' \in M_r : |m - m'| \leq 1\}$ 
18:    $only\_r \leftarrow \{m \in M_r \mid \nexists m' \in M_v : |m - m'| \leq 1\}$ 
19:   if  $|only\_v| > \max(1, 0.10 \cdot |M_v|)$  then
20:      $warnings.APPEND(\text{m/z only in vector})$ 
21:   end if
22:   if  $|only\_r| > \max(1, 0.10 \cdot |M_r|)$  then
23:      $warnings.APPEND(\text{m/z only in raster})$ 
24:   end if
25: end if
26: return  $\{name, peaks, method, warnings\}$ 
```

single peak, and when multiple strokes share a mass the minimum height among them is taken, so that a spurious over-tall stroke (e.g. a gridline or annotation coincident with a bar) cannot inflate the reported intensity.

Label-anchored m/z correction. The tick calibration of Eq. (1) is coarse: ticks are sparsely spaced and each carries sub-pixel position error, so a small slope/intercept drift accumulates across the axis. On some real-instrument exports this manifests as a near-uniform rightward drift of ≈ 0.45 mass units, enough to push affected bars across the 0.5 rounding boundary and assign the wrong integer mass (e.g. m/z 4 $\rightarrow$ 5, 17 $\rightarrow$ 18, 29 $\rightarrow$ 30). Every genuine peak, however, has its integer m/z printed as a text label directly above its bar; these labels are the instrument’s own ground-truth assignment. We recover them from the text layer, keeping only integer words in (1,2000] that sit above the x -axis and horizontally over a bar (rejecting tick labels, the y -axis “100”, and scan

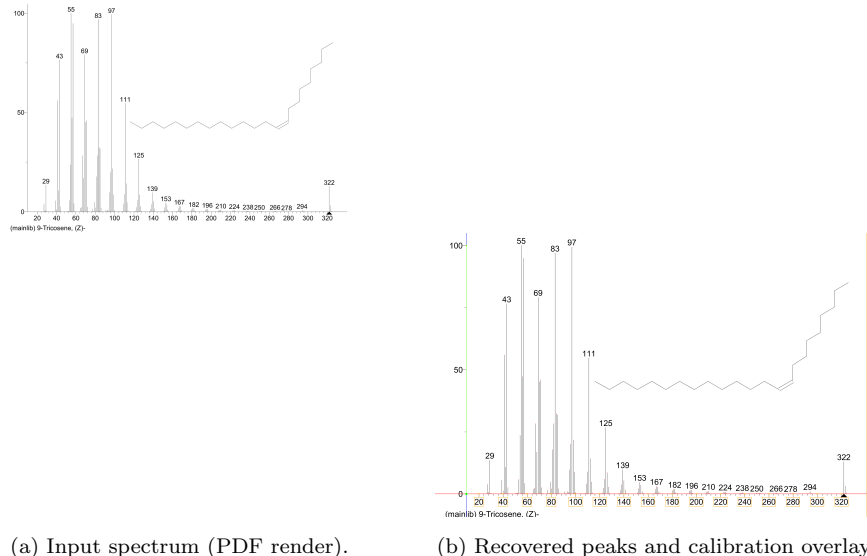


Figure 3: Vector extraction on a representative spectrum (*9-Tricosene, (Z)-*). (a) the input PDF chart; (b) the detected axes, calibration tick labels (here at m/z 20, 40, ..., 320), and the extracted bar tops overlaid on the figure. The recovered peaks coincide with the drawn bars without rasterization.

annotations such as “2.09e9”), and match each to its nearest bar to form anchor pairs (x_j, v_j) of bar abscissa and labelled mass. Rather than a least-squares fit — which a single mislabelled anchor would skew — we fit a robust Theil–Sen line $x = a(m/z) + b$, taking a as the median of all pairwise anchor slopes and $b = \text{median}(x_j - a v_j)$. The estimator is dependency-free, threshold-free, deterministic, and tolerates up to $\approx 29\%$ arbitrary outliers, so a stray label cannot corrupt the assignment. Every bar — labelled or not — is then re-rounded against this clean line, inheriting the corrected slope and intercept. When fewer than three usable anchors are found the fit is left untouched and the original tick calibration is retained with a warning, so the correction never degrades a well-calibrated page. This step is what removes the residual m/z drift observed on the in-house exports; on the NIST corpus, where rendering is clean, anchors and tick calibration already coincide.

Base-peak renormalisation. The “100” label’s text-box centre sits a small, fixed distance above the true 100 % gridline, so Eq. (2) systematically under-reads every bar by a constant offset $\delta = 100 - \max_k I_k$, where I_k are the scaled intensities. We remove it by adding δ back to every peak (clamped at 100), which restores the base peak to exactly 100 % as a mass spectrometer reports it. The correction is a single additive shift applied uniformly to all bars; it is skipped when the maximum already meets 100 %.

3.3 Raster fallback

When no usable vector layer is present, the pipeline falls back to image processing. This section describes that image-based path.

The page is rasterised at 600 DPI — double the typical 300 DPI print resolution — to ensure that adjacent bars remain visually distinct and no peak is lost to sub-pixel mixing. The resulting image is converted to greyscale and binarised using Otsu’s method, which adaptively selects an optimal foreground/background threshold for each document, providing robustness across differently rendered PDFs.

Axis detection proceeds on the binary image by projecting pixel mass onto rows and columns. The x -axis row is the lowest row with a dominant horizontal ink mass; the y -axis column is the leftmost column in the left half of the image with sufficient vertical ink mass. Tick labels are extracted from the PDF text layer and projected into image coordinates to produce the same (x_i, v_i) calibration pairs as the vector pipeline, yielding the least-squares calibration of Eq. (1) and the intensity scaling of Eq. (2).

Bar detection differs from the vector tier, because the raster image has no explicit stroke geometry to read. Rather than scanning for contiguous black runs — which merge adjacent bars into one when the per-mass pixel pitch is small, as it is on wide-range or high- m/z spectra — the scanner samples on the calibrated mass grid: for each integer m/z it computes the expected column, searches a window narrower than half the inter-bar pitch, and records the height of the tallest bar in that window. This gives every integer mass its own reading and prevents touching bars from being collapsed or mis-centred. The scan row is first stepped just above the (several-pixel-thick) baseline so the axis line itself is not read as one continuous bar.

The detected bars then pass through the same post-detection processing as the vector tier without modification: the label-anchored Theil–Sen m/z correction and the base-peak renormalisation of §3.2. Because tick labels come from the PDF text layer, this tier handles figures whose vector drawings are absent but whose text layer survives; PDFs produced by scanning a physical page carry neither layer and are outside the scope of both tiers (see §6.4).

3.4 Validation layer

Extracted peaks are subjected to two layers of automated validation: per-file physics and consistency checks, and a cross-pipeline comparison when both pipelines are run.

Physics and completeness checks. Five invariants are verified against every extracted spectrum, each chosen because its violation maps to a specific, diagnosable extraction failure.

- *Base-peak normalisation.* A mass spectrometer normalises its output so that the tallest bar — the base peak — is reported at 100 %. A maximum intensity below 99.5 % indicates a y -axis calibration failure.

- *m/z range* [1, 2000]. Values outside this range indicate that *x*-axis tick labels were misread during calibration.
- *Intensity range* (0, 100]. Zero or negative values indicate a spurious sub-baseline segment; values above 100 % indicate that the intensity clamp failed.
- *Minimum peak count*. Fewer than three peaks almost certainly indicates that the bar scanner missed most of the chart.
- *Duplicate m/z values*. A unit-resolution instrument produces at most one bar per integer mass; duplicates indicate calibration error of less than 0.5 mass units.

Batch-level outlier detection. Files whose peak count deviates by more than 6σ from the batch mean are flagged for manual inspection, catching systematic failures that pass the per-file thresholds.

Cross-pipeline validation. When cross-validation is enabled, both tiers are run on each file and their outputs compared via Algorithm 1. Two discrepancies trigger a warning: a peak-count difference exceeding 20 % of the larger count, and *m/z* sets differing by more than 10 % after applying a ± 1 mass-unit tolerance.

3.5 Parallel batch driver

Processing a large corpus efficiently requires overlapping disk I/O with CPU computation and minimising serialisation overhead. The driver achieves this through three coordinated design choices.

Sequential byte-preloading. The main thread reads every PDF into memory sequentially before dispatching to workers, amortising filesystem overhead across the whole batch. Workers receive raw bytes and perform no disk I/O themselves.

Bounded-window submission. When a task is submitted to a worker pool it returns a *future*—a lightweight handle that the caller can later query to retrieve the result once the worker finishes. A sliding window of at most N in-flight futures (default $N = 64$) caps peak memory at approximately $N \times \bar{f}$, where $\bar{f}$ is the mean PDF size, while keeping the worker pool continuously saturated.

Single streamed output. All JSON serialisation is performed by the main thread, appending each result as one line to a single open JSONL file. This eliminates the per-file open/allocate/close cycle that dominates wall time at scale, and the output is readable before the batch completes.

Process pools are recommended for all workloads. The benchmark results of §5.5 show that process-level isolation bypasses the GIL and achieves $6.94\times$ speedup ($f_{\text{par}} = 89.7\%$). Thread pools provide no wall-clock reduction on this pipeline: Python result-collection and JSONL serialisation in the main thread remain GIL-bound, regardless of GIL-releasing C extensions in the worker code.

4 Experimental Setup

We evaluate the pipeline on two datasets: a large NIST-derived set for extraction fidelity and a small set of independently produced real-instrument spectra for cross-pipeline consistency on data produced outside the NIST rendering conditions.

4.1 Datasets

Two datasets are used, one for fidelity evaluation and one for cross-pipeline consistency on independently produced data.

NIST. Approximately 50,000 EI-MS reference spectra rendered to PDF, sourced from the NIST MS library [6].

In-house. A collection of real-instrument gas spectra (provided by IISc) with no NIST reference, evaluated by vector-raster agreement; disagreements were checked manually. The collection contains 68 unique spectra used for all reported statistics.

4.2 Matching protocol (NIST)

This subsection describes how each extracted NIST spectrum is paired with its ground-truth library entry and scored.

Reference peak lists were obtained by exporting the m/z -intensity values of all $\sim 267,000$ library compounds from NIST MS Search v2.3. Of 49,351 extracted spectra carrying a parseable compound name, 616 were discarded as duplicate names, leaving 48,576 matched to a NIST library entry and scored by cosine similarity.

4.3 Metrics

We report metrics along two axes: extraction correctness and processing performance.

Cosine similarity. Following standard mass-spectral library search [7], each extracted spectrum is scored against its NIST reference by

$$\cos(a, b) = \frac{\sum_{m \in a \cap b} a_m b_m}{\sqrt{\sum_m a_m^2} \sqrt{\sum_m b_m^2}}. \quad (3)$$

A score near 1.0 reflects faithful recovery of the published peak pattern. This metric is intensity-weighted over shared masses: it measures fidelity to the figure as drawn, not chemical correctness of the underlying spectrum.

Base-peak agreement. A binary per-compound check of whether the extracted base peak (maximum-intensity m/z) coincides with the reference base peak.

Physics pass-rate. The fraction of extracted spectra satisfying all five validation invariants of §3.4.

Parallel performance. For P workers we report speedup and efficiency,

$$S = \frac{T_1}{T_P}, \quad E = \frac{S}{P}, \quad (4)$$

and estimate the parallel fraction f via Amdahl’s law,

$$S(P) = \frac{1}{(1 - f) + f/P}, \quad (5)$$

from which the asymptotic speedup ceiling $1/(1 - f)$ follows. We estimate f by inverting Eq. (5) at the measured endpoint speedup $S(P_{\max})$; the plotted Amdahl curve is therefore a single-point estimate anchored at $P_{\max}$, not a least-squares fit over the worker ladder.

4.4 Hardware and software

All experiments were conducted on a single machine; its specification is given here for reproducibility.

All experiments were run on an Intel Core Ultra 9 185H, with 22 logical cores and 32.0 GB of RAM, under Windows 11, using Python 3.12.10. Key library versions: PyMuPDF (`fitz`) 1.27.2.3 [9], OpenCV 4.13.0, NumPy 2.4.4.

5 Results

This section reports the outcomes of the four evaluations: extraction fidelity on the NIST corpus, pipeline-tier usage, validation-layer pass rates, cross-pipeline consistency on real-instrument data, and parallel performance.

5.1 Fidelity against NIST

The primary correctness evaluation compares each extracted spectrum against its NIST library entry using cosine similarity, as described in §4.3.

Of the 49,351 extracted spectra that carried a parseable compound name, 616 were discarded as duplicate names, leaving 48,576 matched to a NIST library entry. Across these, the pipeline reproduces the reference peak pattern with near-perfect fidelity: 48,574 of 48,576 spectra (99.99 %) attain a cosine similarity of at least 0.99, and the extracted base peak coincides with the reference in the same 48,574 cases.

The distribution is sharply bimodal (Table 1, Fig. 4). No spectrum falls in the intermediate 0.90–0.99 range; every spectrum either matches near-perfectly or fails outright. The two failures are large, structurally complex molecules: *2-Acetamido-5-p-tosylamido-p-benzoquinone* yielded only 2 of its 93 reference peaks (cosine = 0.065), and *Benzo[g]pteridine, 2,4-diamino-6,7,8,9-tetrahydro-7-methyl-* yielded 72 of 125 (cosine = 0.278). Both show base-peak disagreement.

Manual inspection of these PDFs reveals that the mismatch originates in the source files themselves: the PDFs sourced from outside the main NIST rendering pipeline contain an atypical layout that the extractor processed without error, but the peak geometry as drawn does not faithfully represent the underlying library entry. The discrepancy is therefore an artefact of the input PDF, not a failure of the extraction logic. These cases are examined in §6.3.

The NIST PDFs are rendered from the same library used as the reference, so a high cosine measures extraction *fidelity* — whether the drawn geometry is recovered without loss — not generalisation to spectra from other instruments. Cross-pipeline consistency on real-instrument data is assessed in §5.4.

Table 1: Cosine-similarity distribution of 48,576 extracted spectra against the NIST reference. Base-peak m/z agrees in 48,574/48,576 cases (99.99 %).

Cosine range	Count	Fraction
≥ 0.99	48,574	99.99 %
$[0.95, 0.99)$	0	0.00 %
$[0.90, 0.95)$	0	0.00 %
< 0.90	2	0.01 %

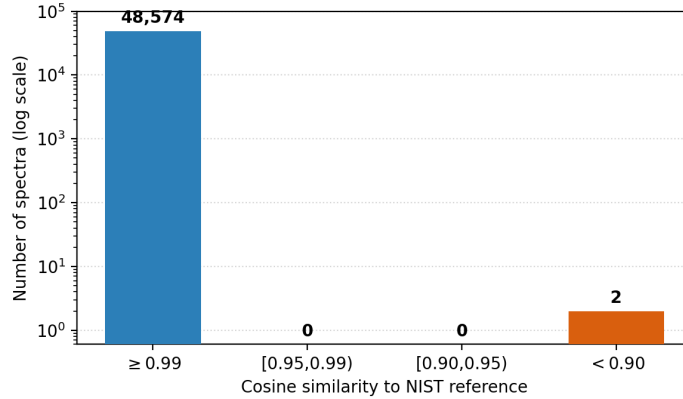


Figure 4: Cosine-similarity distribution of the 48,576 matched spectra (log scale). The result is sharply bimodal: 48,574 spectra reach cosine ≥ 0.99 and only 2 fall below 0.90, with no spectrum in between.

5.2 Tier usage

This subsection records how often each extraction tier was invoked across the full NIST corpus.

Across the full 49,999-file corpus, the vector tier succeeded in 49,970 cases (99.9%); only 29 files (0.1%) required the raster fallback (Table 2). This confirms that standard NIST PDF rendering consistently exposes a usable drawing layer, validating the fast-path design assumption.

Table 2: Pipeline-tier usage across 49,999 NIST PDFs.

Tier	Files	Fraction
Vector	49,970	99.9 %
Raster	29	0.1 %
Total	49,999	100.0 %

5.3 Validation-layer pass rate

Having established extraction fidelity, we report how many files passed the physics-based validation checks described in §3.4.

49,946 of 49,999 files (99.9%) passed every physics check; 33 raised warnings and 20 raised informational alerts; no file failed the schema check (Table 3). The 33 warnings decompose into 32 empty extractions — all 29 raster-fallback files plus 3 vector files in which no bars were detected — and 1 sparse extraction; the 20 alerts are all high-outlier peak counts ($z > 6.0$), all passing physics checks. The mean peak count is 107.1 ($\sigma = 64.2$; range 0–597) and the mean highest m/z is 283.9 (range 18–1328).

Table 3: Validation-layer outcome for 49,999 extracted spectra.

Outcome	Files	Fraction
Clean	49,946	99.9 %
Warning (action required)	33	0.1 %
Alert (informational)	20	0.0 %
Failed (schema error)	0	0.0 %

5.4 Cross-pipeline consistency on real-instrument data

The in-house set carries no external reference, so it cannot measure extraction accuracy directly; what it provides is a check of whether the two independent extraction paths — vector and raster — converge on the same peak list for data produced outside the NIST rendering conditions.

A prerequisite for a meaningful comparison is correct m/z assignment. Unlike the NIST PDFs, which are rendered under controlled conditions, the in-house exports carry a near-uniform ≈ 0.45 -unit rightward drift in the tick axis. Left uncorrected, this drift pushes several bars across the 0.5 rounding boundary

and assigns the wrong integer mass (e.g. m/z 4 $\rightarrow$ 5, 17 $\rightarrow$ 18), which would cause apparent disagreement between the two tiers even when both correctly read the same drawn bar. The label-anchored Theil–Sen correction of §3.2 eliminates this drift by fitting a robust regression line through bar-position–label anchor pairs and re-rounding every bar against its corrected abscissa. It is this correction that makes the cross-pipeline comparison meaningful for the in-house data; without it, the m/z misassignment would artificially suppress cosine similarity regardless of how accurately each tier detected the bar heights.

Both tiers were run on the 68 unique in-house gas spectra after applying the correction, and compared by cosine (Eq. 3). The mean agreement is 0.9963 and 63 of 68 spectra (92.6 %) achieve ≥ 0.99 (Table 4); base-peak m/z coincides in all 68/68 (100.0 %). The distribution is slightly broader than the NIST result, reflecting greater variability in rendering across independent instrument software, but no file falls below 0.90. Because both tiers share the Theil–Sen correction, their agreement reflects consistency of bar detection rather than of calibration. The lowest-agreement files are *H2_70%-CO_15%-CO2_15%_03.pdf* (cosine = 0.9287, BP+) and the *Methane_UHP* spectra (cosine = 0.9363, BP+); these are characterised in §6.3.

Table 4: Vector–raster cosine distribution for the 68 in-house spectra. Mean cosine = 0.9963; base-peak agrees in 68/68 (100.0 %).

Cosine range	Count	Fraction
≥ 0.99	63	92.6 %
[0.95, 0.99)	2	2.9 %
[0.90, 0.95)	3	4.4 %
< 0.90	0	0.0 %

As a complementary large-scale consistency check, both tiers were also run on the full $\sim 50,000$ -file NIST corpus. Of the 50,046 files for which both tiers produced non-empty output, 50,045 ($> 99.99\%$) agree at cosine ≥ 0.99 (mean 1.0000), with base-peak m/z agreement in 50,019/50,078 (99.9 %). Unlike the in-house set — where every file exercises the raster tier — the great majority of corpus files are handled natively by the vector tier, so this measures internal consistency between the two paths on clean renders rather than generalisation; the near-unanimous agreement confirms that the raster fallback reproduces the vector result wherever both apply.

5.5 Parallel performance

This subsection evaluates how well the parallel batch driver scales with worker count, comparing thread and process pools.

The batch driver was benchmarked on 17,068 PDFs (corpus folders F1–F17 plus the in-house gas set) across $P \in \{1, 2, 4, 8, 11, 16, 22\}$ workers on a 22-logical-core machine (Windows 11, Python 3.12.10), three trials per configura-

tion, comparing thread and process pools (Table 5, Fig. 5). This subset keeps the full sweep (two pools $\times$ seven worker counts $\times$ three trials) tractable; because wall time is dominated by per-file parsing, which is independent of corpus size, the per-worker scaling it measures carries over to the full corpus.

The thread pool achieves no meaningful speedup: wall time rises slightly from 129.2 s at $P = 1$ to 148.5 s at $P = 22$ (speedup $1.00\times\text{--}0.87\times$, $f_{\text{par}} \approx 0\%$). Although PyMuPDF, OpenCV, and NumPy release the GIL during C-level work, Python result-collection and JSONL output retain it, serialising these phases under all worker counts.

The process pool bypasses the GIL entirely: wall time falls from 129.3 s at $P = 1$ to 18.6 s at $P = 22$ ($6.94\times$, 917 PDFs/s), with $f_{\text{par}} = 89.7\%$ (serial fraction 10.3%) and Amdahl ceiling $9.7\times$. The $P = 2$ configuration achieves 102.7% efficiency (superlinear), a process-startup cache-warming artefact. The marginal-return sweet spot is $P = 2\text{--}4$ (89.9%–102.7% efficiency); absolute speedup continues rising through $P = 22$.

Table 5: Thread vs. process pool on 17,068 PDFs, three-trial average (thread $T_1 = 129.2$ s; process $T_1 = 129.3$ s). The process pool reaches $6.94\times$ ($f_{\text{par}} = 89.7\%$); the thread pool shows no speedup ($f_{\text{par}} \approx 0\%$).

P	Thread pool		Process pool	
	T_p	S	T_p	S
1	129.2 s	1.00 $\times$	129.3 s	1.00 $\times$
2	131.1 s	0.99 $\times$	62.9 s	2.05 $\times$
4	137.2 s	0.94 $\times$	35.9 s	3.60 $\times$
8	140.7 s	0.92 $\times$	25.1 s	5.16 $\times$
11	143.3 s	0.90 $\times$	22.2 s	5.83 $\times$
16	145.8 s	0.89 $\times$	19.5 s	6.62 $\times$
22	148.5 s	0.87 $\times$	18.6 s	6.94 $\times$
f_{par}	$\approx 0\%$		89.7%	
Ceiling	1.0 $\times$		9.7 $\times$	

6 Discussion

The results presented in §5 raise several questions of interpretation: why the vector-first approach achieves near-lossless fidelity, what the residual failures reveal about the pipeline’s limits, and where the method will not generalise. These are addressed in turn below.

6.1 Why the vector-first design works

This subsection explains why reading the PDF vector layer directly produces higher fidelity than image-based approaches on the same data.

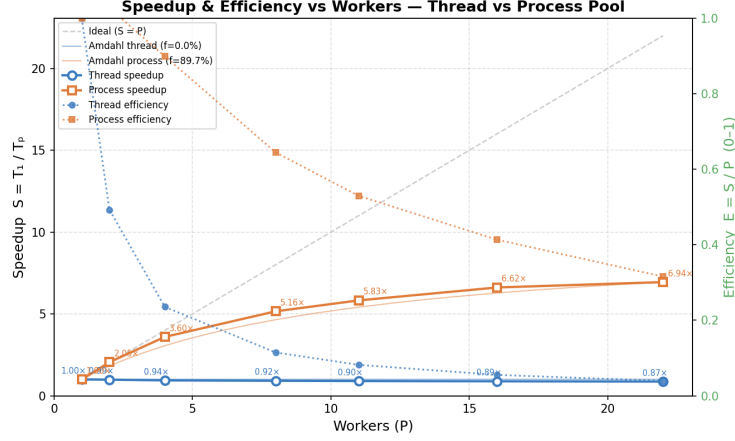


Figure 5: Measured speedup versus worker count for the process and thread pools on 17,068 PDFs (three-trial average), with the Amdahl curve (parameter f estimated from the endpoint speedup, not least-squares fitted). The process pool scales to $6.94\times$ at $P = 22$ ($f_{\text{par}} = 89.7\%$, ceiling $9.7\times$); the thread pool stays flat near $1\times$ because the GIL serialises result-collection and JSONL output.

A PDF drawing layer is not an approximation of a spectrum — it is the exact sequence of coordinates the rendering software used to place each bar. Reading it recovers bar positions at floating-point precision: there is no pixel grid to quantize against, no threshold to tune, and no learned detector to fail on unfamiliar rendering styles. When calibration tick labels are also present in the text layer, the m/z -pixel mapping is a linear fit to exact positions rather than to blob centroids. The round-trip error reduces to floating-point rounding in the least-squares fit, which is negligible at the scale of integer mass units.

The corpus validates this at scale. The vector tier handled 49,970 of 49,999 files (99.9%), and among these 48,574 of 48,576 matched spectra reached cosine ≥ 0.99 with base-peak agreement. The two failures are missed-extraction events — the calibration was applied correctly, but a small number of bars were not detected — rather than calibration errors that would shift every peak proportionally. The absence of any cosine in the 0.90–0.99 range confirms that the pipeline either recovers a spectrum faithfully or produces a sparse result that is identifiably incomplete.

The parallel performance result reinforces the same independence principle. Because each file is processed without shared state, the work is embarrassingly parallel; process-level isolation removes the GIL as a serialisation bottleneck and achieves $6.94\times$ speedup with an 89.7% parallel fraction. The 10.3% serial residual reflects the minimum coordination cost — sequential preloading and single-writer output — that is inherent to any corpus-scale pipeline.

6.2 Ablations and sensitivity

Formal component-isolation ablations were not conducted because the near-perfect NIST result leaves little variance to attribute. Each component’s contribution is instead characterised through two indirect sources of evidence.

The validation layer provides per-check coverage: zero spectra triggered a physics failure across 49,999 files, confirming that calibration produced no out-of-range outputs. Of the 32 empty-extraction warnings, all 29 raster-fallback files came up empty — showing that the raster tier is invoked precisely on files the vector tier cannot handle — and the remaining 3 are vector files in which no bars were detected. The 20 high-outlier alerts (all passing physics checks) represent genuinely dense spectra, confirming that the 6σ threshold correctly identifies statistical outliers.

The in-house cross-validation serves as an indirect sensitivity test of the raster tier. A mean vector–raster cosine of 0.9963 across 68 unique independently produced files (68/68 base-peak agreement) shows that the two tiers converge on the same peak list in almost all cases, with disagreements being diagnosable rather than random.

6.3 Failure analysis

The following examines the small number of cases where the pipeline produced a low-agreement result, separating failures attributable to the source data from those attributable to extraction behaviour.

NIST failures. Both failure cases originate in the source PDF, not in the extraction logic. The charts embedded in these files are themselves incomplete: they do not render the full set of peaks recorded in the NIST library entry. The extractor correctly reads what is drawn — calibration is accurate and recovered peaks sit at the correct m/z values — but because the chart omits most of the spectrum, the extracted peak list is necessarily sparse. *2-Acetamido-5-p-tosylamido-p-benzoquinone* has only 2 bars present in the PDF against 93 in the reference (cosine = 0.065, BP–); *Benzo[g]pteridine, 2,4-diamino-6,7,8,9-tetrahydro-7-methyl-* has 72 bars against 125 (cosine = 0.278, BP–). The low cosine similarity is the expected outcome when the input chart is incomplete, and it is the correct behaviour of the pipeline: a faithful extractor cannot recover peaks that were never drawn. Both files are correctly flagged by the validation layer (Fig. 6).

In-house disagreements. All 68 unique files agree on the base peak and none falls below cosine 0.90; the residual disagreements are confined to low-intensity peaks that one tier detects and the other misses. The lowest-agreement case, *H2_70%-CO_15%-CO2_15%-03.pdf* (cosine = 0.9287, vector 20 peaks, raster 16 peaks, BP+), and the *Methane_UHP* spectra (cosine = 0.9363, vector 20 peaks, raster 16, BP+) both arise from the vector tier recovering small fragment peaks the raster grid-sampler misses: for methane, vector resolves m/z 14 (CH_2^+) at 67.7% relative intensity, which the raster tier drops, accounting for almost the entire cosine gap. Because the vector layer encodes

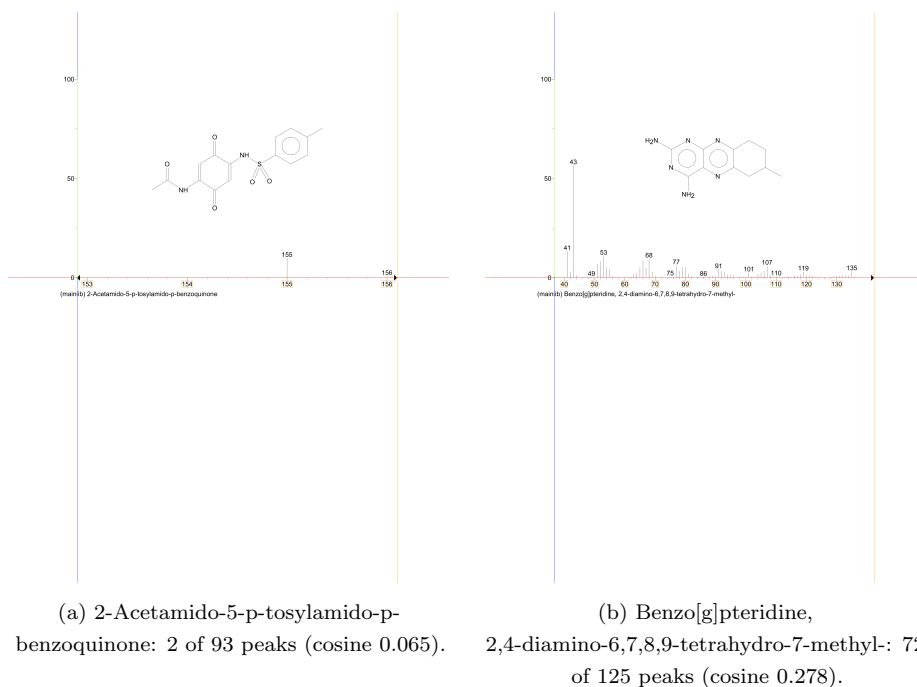


Figure 6: The two NIST failure cases, shown with the extraction overlay. In both, the detected axes and calibration are correct and the extractor faithfully reads what is drawn; the low cosine similarity arises because the source PDF chart is itself incomplete, omitting most peaks present in the NIST library entry. Both are correctly flagged by the validation layer.

the bar exactly, these cases are evidence of vector *recovering* information that rasterization discards, rather than of an extraction error.

6.4 Limitations

No extraction pipeline is universal. This subsection identifies the conditions under which the current approach will degrade or fail, and suggests directions for future work to address them.

Requires a vector or text layer. PDFs produced by scanning carry neither vector drawing operators nor a selectable text layer; the raster path can detect bars but calibration fails without legible tick labels. The 29 raster-fallback empty extractions in the corpus illustrate this boundary. Future work should incorporate an OCR-based tick-label path.

Assumes unit-resolution integer m/z . The calibration rounds every abscissa to the nearest integer, which is correct for standard GC-MS but incorrect for high-resolution instruments (Orbitrap, Q-TOF) reporting accurate masses to multiple decimal places.

Single-spectrum-per-page assumption. Axis detection selects the longest near-horizontal and near-vertical strokes on the page. Multi-panel figures containing stacked spectra would cause the detector to span multiple plots.

Fidelity $\neq$ generalisation at scale. The 99.99 % NIST result measures round-trip fidelity under controlled rendering. The in-house experiment (68 unique files, mean cosine 0.9963) provides cross-pipeline evidence on independently produced spectra but, lacking an external reference and at a sample size modest relative to the diversity of GC-MS instruments and reporting conventions in the published literature, is not a measurement of accuracy.

7 Conclusion

We have presented a two-tier pipeline for extracting m/z -intensity peak lists from bar mass spectra in PDF documents. By reading the PDF vector drawing layer directly, the common case requires no rasterization, no threshold tuning, and no learned components, yielding near-lossless round-trip fidelity across 48,576 NIST spectra (99.99 % at cosine ≥ 0.99) and a mean vector-raster agreement cosine of 0.9963 on 68 independently produced real-instrument spectra (68/68 base-peak agreement). A process-pool parallel driver achieves $6.94\times$ speedup (917 PDFs/s, Amdahl ceiling $9.7\times$) by exploiting the embarrassing parallelism of per-file independent extraction. The two observed NIST failures — both densely packed large-molecule spectra — are identifiable by the built-in validation layer and represent a well-defined failure mode amenable to targeted improvement.

Future directions include: an OCR-based tick-label extraction path to handle scanned documents outside both current tiers; sub-unit m/z resolution to support high-accuracy instruments; multi-panel figure detection for comparison figures in the literature; and evaluation on a larger, more diverse sample of third-party PDFs to characterise generalisation at publication scale. Beyond extraction itself, the large machine-readable spectral datasets that this pipeline enables are a natural input to the machine-learning models for compound identification and spectrum prediction that currently face a training-data bottleneck.

Data availability

The source code for the extraction pipeline, validation layer, and benchmark driver is available at [REPOSITORYURL---TOFILL]. The NIST EI-MS spectra used for the fidelity evaluation are derived from the NIST/EPA/NIH Mass Spectral Library [6], which is distributed under a commercial licence and cannot be redistributed; they are available to licence holders from NIST. The real-instrument gas spectra are available at the CGPL Department, Ilsc.

Author contributions

Conflicts of interest

There are no conflicts to declare.

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